

TCS NQT: Mensuration & Geometry Masterclass

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1. The Examiner's Cheat Sheet: Concepts & Shortcuts

Golden Rules for Exam Success

- **Volume Conservation:** When melting one shape into another, always equate Volumes: $V_1 = V_2$. Never equate Surface Area.
- **Smart Scaling:** If dimensions are scaled by 'k', Perimeter scales by 'k', Area by ' k^2 ', and Volume by ' k^3 '.
- **Circle/Square Property:** For a given perimeter, the Circle always encloses the maximum area. For a given area, the Square has the minimum perimeter.
- **Inscribed/Circumscribed:**
 - Square inside Circle (Radius R): Diagonal of square = $2R$.
 - Circle inside Square (Side a): Diameter = a.

2. Examiner's Trap Zone

Trap 1: Diameter vs. Radius

TCS questions often provide Diameter but formulas require Radius ($r = d/2$). Always verify what the question asks for (Area vs. Diameter).

Trap 2: Curved vs. Total Surface Area

For a cylinder, open pipes have CSA ($2\pi rh$), while closed cans require TSA ($2\pi rh + 2\pi r^2$). Misreading "open tank" vs "closed tank" is a guaranteed point loss.

Trap 3: Unit Mismatch

Mixing cm and meters is the #1 reason for incorrect answers. Convert all dimensions to the same unit *before* starting your calculation.

3. 10 High-Priority Questions

Q1: (Volume Conservation) A hollow spherical ball with internal radius 1 cm and external radius 5 cm is melted into 61 identical small solid spherical balls. Find the diameter of each small ball.

Sol: $V_{\text{hollow}} = \frac{4}{3} \pi (R^3 - r^3) = \frac{4}{3} \pi (125 - 1) = \frac{4}{3} \pi * 124$. $61 * (\frac{4}{3} \pi * r^3) = \frac{4}{3} \pi * 124$. $r^3 = 124/61 \approx 2$. $r = 1.03$ cm. Diameter ≈ 2 cm.

Q2: (Perimeter comparison) 6 square plots of side 'a' are connected end to end to form a rectangle of area 726 m². Compare the perimeter to a circle of radius 10m.

Sol: $6 * a^2 = 726 \Rightarrow a^2 = 121 \Rightarrow a = 11$. Rectangle dimensions = 11 x 66. Perimeter = $2*(11+66) = 154$. Circle circumference = $2 * \frac{22}{7} * 10 = 62.8$. Ratio = $154 / 62.8 \approx 2.45$.

Q3: A rectangular plot has length 20m and breadth 15m. Find the area of a 2m wide path built inside the perimeter.

Sol: Area = $(20*15) - (20-4)*(15-4) = 300 - (16*11) = 300 - 176 = 124$ m².

Q4: The diagonal of a square is $10\sqrt{2}$ cm. Find its area.

Sol: Diagonal = $a\sqrt{2} \Rightarrow a=10$. Area = $a^2 = 100$ cm².

Q5: If the radius of a cylinder is doubled and height is halved, what happens to the volume?

Sol: $V = \pi r^2 h$. New $V = \pi (2r)^2 (h/2) = \pi (4r^2) (h/2) = 2\pi r^2 h$. The volume doubles.

Q6: Find the area of an equilateral triangle with side 6 cm.

Sol: Area = $(\sqrt{3}/4) * a^2 = (\sqrt{3}/4) * 36 = 9\sqrt{3}$ cm².

Q7: A room is 10m long, 8m wide, and 4m high. Find the area of the 4 walls.

Sol: Area = $2*(l+b)*h = 2*(10+8)*4 = 2*18*4 = 144$ m².

Q8: Two cylinders have radii in ratio 2:3 and heights in ratio 5:3. Find the ratio of volumes.

Sol: $V_1/V_2 = (r_1^2/r_2^2) * (h_1/h_2) = (4/9) * (5/3) = 20/27$. Ratio = 20:27.

Q9: A circular wire of radius 14 cm is bent into a square. Find the side of the square.

Sol: Circumference = $2 * 22/7 * 14 = 88$. Perimeter of square = $4a = 88 \Rightarrow a = 22$ cm.

Q10: The volume of a cube is 216 cm^3 . Find its total surface area.

Sol: $a^3 = 216 \Rightarrow a = 6$. TSA = $6a^2 = 6 * 36 = 216 \text{ cm}^2$.